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Abstract

A good abstract is a stand-alone summary of the paper, and should summarize the key components of the manuscript. Generally, the abstract should be concise and informative within 200-300 words. As an abstract is a separate section, it should be a self-containing text (no abbreviations, no references, no URLs, no undefined concepts, etc.). For research articles, abstracts should give a pertinent overview of the work. We strongly encourage authors to use the following style of structured abstracts, but without headings: (1) Background: Place the question addressed in a broad context (research background) and highlight the purpose of the study; (2) Methods: Describe briefly the main methods or treatments applied; (3) Results: Summarize the article's main findings; and (4) Conclusions: Indicate the main conclusions or interpretations.

Keywords: cybersecurity, information security, threat analysis, resilience

Chapter 1

Introduction

1.1 Background

The Internet of Things (IoT) connects physical objects, sensors, and various devices through communication networks, enabling data-driven services in domains such as healthcare, industrial automation, and smart environments. Alongside the expansion of IoT deployments, the number of connected components exposed to cybersecurity threats increases correspondingly [1]. The Message Queuing Telemetry Transport (MQTT) protocol is commonly used for lightweight publish-subscribe communication in IoT environments. However, MQTT-based systems can become targets for denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks. The DoS/DDoS-MQTT-IoT dataset, collected from a physical IoT testbed, provides normal MQTT traffic alongside ten DoS attack scenarios to facilitate the study of attacks in this context [2]. Intrusion Detection Systems (IDS) monitor network activity to identify traffic or behavior exhibiting malicious signs. Machine learning and deep learning methods have been utilized and researched for this task due to their ability to learn characteristic patterns directly from network traffic. Their effectiveness depends partly on the availability and representativeness of the training data [3]. Federated Learning (FL) allows multiple clients to collaboratively train a shared model while keeping their data local; a central server distributes the global model and aggregates updates from clients across training rounds [4]. This approach is highly suitable when IoT data is distributed across multiple devices or locations, although communication and computational costs remain critical factors to consider in resource-constrained environments [5, 6]. Keeping data local also does not equate to completely eliminating privacy concerns arising from the exchange of model updates [7].

Privacy-preserving synthetic data generation is a potential solution to address the shortage of training data for rare attack classes in IoT networks while still ensuring strict security

guarantees. Rather than relying on deep neural networks that require substantial computational resources and face convergence difficulties on edge devices [8], the probabilistic graphical model paradigm (Probabilistic Graphical Models) approaches the problem by decomposing tabular data into low-dimensional marginal frequency distribution slices (low-dimensional marginals) [9]. Through the Chow-Liu dependency tree structure (Chow-Liu Tree) [10], this approach effectively captures the core correlations between traffic features and attack labels without being affected by the curse of dimensionality. When combined with Differential Privacy (DP) [9] on contingency tables and consistent graphical inference mechanisms (such as Private-PGM), the system can reconstruct a valid global distribution to generate conditional traffic data, establishing a foundation to enhance the performance of downstream intrusion detection systems (IDS) [11, 12].

1.2 Problem Statement

Building an effective IDS for IoT networks faces multiple challenges related to data privacy, the representation of attack classes, and the practical costs of the model training process. These challenges are particularly evident when training data is distributed among clients and synthetic attack samples are considered as a supplementary detection solution. The first problem relates to privacy risks during federated training. Although FL avoids the periodic transfer of raw local data, clients must still exchange model updates with the server. Previous studies have demonstrated that shared gradients can leak information about training data in certain attack scenarios [7]. Therefore, sharing model weights instead of raw data cannot be considered a comprehensive privacy protection measure. Depending on the threat model and the type of information to be protected, supplementary mechanisms may be mandatory. The second problem concerns the representation of attack classes in the training data. Class imbalance can bias the learning process towards majority classes, making it difficult to identify classes that account for a low percentage [3]. For the DoS/DDoS detection problem in MQTT, the impact of this issue depends on the dataset version, label mapping, preprocessing pipeline, and data partitioning strategy for clients. Therefore, the class distribution of the dataset after selection and preprocessing must be empirically verified rather than assumed by default. If some attack classes lack data, synthetic samples can supplement training examples. However, merely increasing the number of samples does not prove that the generated data is useful; their actual contribution must be evaluated through data quality metrics and the detection

performance of the downstream IDS model.

The third challenge pertains to the hardware constraints of edge devices, the "curse of dimensionality," and the intricate trade-off among privacy protection, data quality, and operational overhead: on the one hand, deploying traditional distributed deep generative models (such as GANs or CVAEs) requires substantial computational resources (GPU/CPU) and large RAM capacity at client nodes, which is infeasible for hardware-constrained edge IoT devices [8]. On the other hand, tabular MQTT network traffic inherently possesses a high-dimensional structure with numerous interdependent attributes. Applying Differential Privacy (DP) directly to this high-dimensional attribute space readily leads to a state-space explosion—reaching up to 10^{19} cells [11]—causing the injected differential privacy noise to completely overwhelm the original data signal and severely degrade the fidelity of the generated distributions [9].

A direct consequence is a severe tug-of-war among three factors: the degree of privacy protection (privacy budget ε), synthetic data utility for downstream IDS models, and computational/operational overhead across the IoT infrastructure. Specifically, tightening privacy guarantees (reducing the budget ε) injects heavy DP noise that destroys the core feature correlations of rare DoS/DDoS attack classes [9, 12]. Conversely, increasing the privacy budget and expanding model capacity to improve the quality of generated data causes computational costs and system runtime to surge drastically [12]. Consequently, finding a tabular data generation solution that simultaneously ensures rigorous privacy guarantees, remains computationally lightweight for edge devices, and preserves critical feature characteristics for rare attack classes remains a major, unresolved technical bottleneck.

1.3 Research Objectives and Research Questions

1.3.1 Objectives

This study proposes PP-FedData—a privacy-preserving federated data generation framework for intrusion detection systems (IDS) in IoT-MQTT networks. The research aims to achieve four primary objectives (O1–O4):

- **Objective O1 – Federated Architecture Construction:** Model each IoT Gateway device as a client within the Federated Learning network (FL Client). Each gateway

stores raw packet traffic data locally and only transmits protected statistics to the Aggregation Server under an honest-but-curious threat model.

- **Objective O2 – Marginal-based Federated Generator Development (FedDP-Marginal):** Extend marginal-based privacy-preserving data synthesis methods (such as MST and PrivBayes) to the federated learning environment: each gateway releases noisy per-class count tables across three communication rounds. Noise can be partitioned among gateways and accumulated via the Secure Aggregation mechanism (SecAgg$^+$) (corresponding to Distributed Differential Privacy using discrete Skellam noise), or fully added locally at each gateway (Local DP). Synthetic data records are subsequently sampled from a Chow-Liu dependency tree structure trained on the released marginal statistics tables.

- **Objective O3 – Deep Generative Baseline Implementation:** Implement a federated Conditional Variational Autoencoder (CVAE) model as a deep generative baseline. The CVAE model is trained using the FedAvg algorithm and protected by DP-SGD, SecAgg$^+$, or a combination of both, under identical data partitioning and evaluation protocols to enable a head-to-head empirical comparison against the FedDP-Marginal generator.

- **Objective O4 – Comprehensive Evaluation and Deployment Guidance Formulation:** Comprehensively evaluate all privacy-preserving configurations on a held-out real test set across five dimensions (utility, fidelity, privacy, overhead, and statistical rigor). From these empirical findings, establish a transparent decision rule to map practical deployment constraints (server/client trust assumptions, privacy budget ε, communication bandwidth, prioritized attack classes) into their corresponding optimal configurations.

1.3.2 Research Questions

To fulfill the four aforementioned objectives, this research focuses on addressing four core research questions:

- **RQ1:** When raw network traffic data cannot be centralized, how much IDS utility does the synthetic data generated by the federated generator retain compared to an IDS trained on centralized real data? Does augmenting training sets with synthetic data

improve detection performance when real data is already available (either centrally or at individual gateways)?

- **RQ2:** Under the same privacy budget (ε, δ), how does the federated marginal-based generator (FedDP-Marginal) compare to the federated CVAE model trained with DP-SGD regarding downstream utility, detection capabilities on rare attack classes, and network communication overhead?
- **RQ3:** What overhead does each protection mechanism incur under DP, SecAgg$^+$, or their combination, and at what granularity does it provide privacy protection? How do the unit of privacy protection (packet, TCP stream, or capture session) and the number of honest gateways alter privacy guarantees and operational costs?
- **RQ4:** Which protection configuration (combining DP, SecAgg$^+$, noise distribution, and privacy protection unit) should be selected for specific real-world deployment requirements?

1.4 Significance of the Study

This study addresses two fundamental bottlenecks in collaborative intrusion detection across IoT-MQTT networks: (1) rare attack classes are sparsely and non-identically distributed across IoT Gateway devices (Non-IID), and (2) gateways cannot share raw packet traffic data due to privacy concerns.

The study offers three primary scientific and practical contributions:

- **Marginal-Based Distributed Generator (FedDP-Marginal):** Proposes a distributed synthetic data generation solution in which marginal statistics released by individual gateways are simultaneously protected by formal Differential Privacy (DP) and Secure Aggregation (SecAgg$^+$).
- **Controlled Benchmark:** Conducts a direct, head-to-head empirical comparison between the marginal-based generator (FedDP-Marginal) and the deep generative baseline (Federated CVAE) under identical data partitioning schemes, consistent privacy-preserving mechanisms, and a rigorous evaluation protocol.
- **Requirement-Driven Configuration Guide:** Proposes a transparent decision guideline to assist IoT system administrators in selecting the most suitable privacy configuration

based on their specific threat model, mutual trust assumptions, and practical resource budgets.

Explicitly Stated Limitations and Privacy Guarantees:

The privacy guarantees in this study are precisely and transparently defined:

- **Differential Privacy (DP)** provides mathematical upper bounds limiting the amount of information that released statistical tables or models can leak regarding a single data record (by default at the packet level, or optionally at the TCP stream or capture session level under specific assumptions). The DP mechanism does not anonymize class labels and does not enforce privacy guarantees on the hyperparameter tuning process conducted on validation data.
- **Secure Aggregation (SecAgg$^+$)** only guarantees the confidentiality of individual gateway updates/statistics with respect to the Aggregation Server; it does not prevent what the final model or generated synthetic data might leak about the aggregate dataset.

1.5 Scope and Limitations

This research is scoped within the following specific experimental and technical boundaries:

- **Dataset and Data Granularity:** Utilizes the DoS/DDoS-MQTT-IoT dataset at the packet level (where each record corresponds to a single network packet and includes six traffic class labels: one Normal class and five DoS/DDoS attack families).
- **Federated Simulation Architecture:** A network of five simulated IoT Gateway devices is deployed under an uneven data distribution across nodes (Dirichlet non-IID partitioning).
- **Generative Models and Privacy Mechanisms:** Evaluates and compares two generators: the primary marginal-based federated generator (FedDP-Marginal) and the deep generative baseline (Federated CVAE). Enforces Differential Privacy (DP) across privacy budget levels $\varepsilon \in \{1, 5, 10\}$ with $\delta = 10^{-5}$, Secure Aggregation (SecAgg$^+$), as well as a hybrid configuration combining both mechanisms.

- **Experimental Infrastructure:** The entire federated training workflow is simulated on a single physical host machine using the Flower framework.

Despite achieving rigorous empirical findings, this study is subject to several scientific limitations:

- **Single-Dataset Reliance:** All empirical evaluation results and configuration selection guidelines are derived exclusively from the DoS/DDoS-MQTT-IoT dataset; therefore, generalizability to other IoT application protocols (such as CoAP or HTTP/2) requires further empirical validation.
- **Packet-Level Independence Assumption:** Packets are modeled independently as tabular records, which does not fully capture the temporal/sequential dependencies inherent in an end-to-end communication session or stream.
- **Default Record-Level Privacy Scope:** The Differential Privacy (DP) guarantee applies by default at the individual record level (a single packet). Extending protection to an entire TCP stream or a full packet capture session represents an advanced configuration option that necessitates a trade-off resulting in degraded synthetic data utility.
- **Privacy Budget ε Exclusions:** The privacy budget ε covers only the released count statistics and trained models. Auxiliary metadata such as attack class labels, global preprocessing statistics, and validation-based hyperparameter tuning fall outside the theoretical guarantee of ε.
- **Simulation Environment:** Conducting the simulation on a centralized physical machine does not fully capture real-world network latency, packet loss rates, or the physical hardware constraints of resource-limited edge IoT devices.
- **Constrained Threat Model:** The study operates under an honest-but-curious threat model across all gateways. Sophisticated adversarial threat scenarios such as data/model poisoning attacks or Byzantine clients are outside the scope of this investigation.

1.6 Thesis Structure

The remainder of this thesis is structured into six chapters as follows:

- **Chapter 1: Introduction:** Outlines the background, problem, objectives, significance, scope, and structure of the study.
- **Chapter 2: Literature Review:** Reviews relevant studies and identifies the research gap and the position of this study in the existing literature.
- **Chapter 3: Methodology:** Details the research design, the dataset and its preparation, the two generators, the privacy mechanisms and accounting, and the evaluation protocol.
- **Chapter 4: Experimental and Results:** Describes the experimental setup and reports synthetic-data, IDS, and system overhead results.
- **Chapter 5: Discussion:** Interprets the findings, examines the observed trade-offs, and discusses the limitations and implications of the study.
- **Chapter 6: Conclusion and Future Work:** Summarizes the conclusions and outlines directions for further research.

Chapter 2

Literature Review

Reviews relevant existing research and demonstrates how your thesis fits into the broader academic conversation.

2.1 Review of Previous Studies

2.1.1 Overview of Privacy-Preserving Mechanisms in Federated Learning

Overall Landscape and Scope of Research

The integration of Federated Learning (FL) into Intrusion Detection Systems (IDS) for IoT environments has resolved the issue of raw data centralization, yet the model weight sharing process itself still harbors data leakage risks [13]. To counter this, current studies classify Privacy-Preserving Federated Learning (PPFL) techniques into five main categories: Differential Privacy (DP), Homomorphic Encryption (HE), Secure Multi-Party Computation (SMPC), Trusted Execution Environments (TEE), and Blockchain [13]. Although TEEs provide a secure execution environment and Blockchain ensures transparency, they require specialized hardware or complex consensus mechanisms, which are unsuitable for the constrained resources of edge/IoT devices [14]. Therefore, this study implements DP and SMPC-based secure aggregation (SecAgg$^+$); HE is reviewed as an alternative but not implemented, because of its computational cost on constrained devices [15].

Trusted Execution Environment (TEE)

The Trusted Execution Environment (TEE) provides a completely isolated and secure hardware computing area for processing sensitive data, effectively preserving privacy.

However, this technology is heavily dependent on specialized hardware security mechanisms and verifiable components. This strict hardware requirement makes TEE cumbersome and unsuitable for deployment across lightweight IoT networks comprising massively resource-constrained devices [13].

Blockchain

Blockchain preserves data privacy through decentralized ledgers and cryptographic algorithms that ensure tamper-resistance and eliminate the need for intermediaries. However, this system suffers from fatal performance weaknesses due to low throughput, high latency, and expensive verification costs stemming from continuously growing immutable storage. These severe latency and massive storage overheads make blockchain highly impractical for lightweight, multi-device IoT environments that demand rapid response times [13].

Homomorphic Encryption (HE)

HE enables computations to be performed directly on encrypted data, ensuring absolute security for model updates from clients before server aggregation without the need for decryption [15]. However, the computational cost and communication requirements of HE algorithms are extremely immense and heavy, leading to a surge in system latency [15]. This renders HE less feasible for systems requiring real-time responses and operating under resource constraints, such as IoT-based IDS [15].

Differential Privacy (DP)

Unlike the cumbersomeness of HE, Differential Privacy (DP) offers excellent scalability. The core principle of DP is to mask the contribution of any single data point by adding random noise to the model update at each client [16]. DP is exceptionally lightweight, consumes no extra bandwidth, and easily scales to tens of thousands of participating clients [16]. The tradeoff of this method is that injecting noise into the model inevitably degrades data accuracy, forcing engineers to constantly seek an optimal balance for the privacy budget [16].

Secure Multi-Party Computation (SMPC/SecAgg)

Conversely, Secure Aggregation (SecAgg) protocols based on SMPC comprehensively resolve the weakness of DP. SMPC prevents the server from directly observing individual client updates by ensuring that the server only receives the aggregated model update [17]. Secure aggregation adds no statistical noise; with the fixed-point quantisation used by SecAgg$^+$, the aggregate differs from the plain sum only by a bounded quantisation error [18]. As a trade-off, the original Secure Aggregation protocol requires multiple rounds of complex interactions between clients to exchange secret shares, causing severe bandwidth bottlenecks as the client population grows [19]; however, the subsequent SecAgg$^+$ protocol successfully resolves these overhead issues by replacing the complete communication graph with a k-regular graph, thereby reducing the per-client communication and computation complexity to a polylogarithmic scale [18].

Hybrid Privacy-Preserving Systems

The study by Gutti et al. [20] (integrating DP and SecAgg via the HADA algorithm) demonstrates that combining DP (to prevent inference attacks) and SecAgg (to protect the transmission channel) is a promising trajectory. However, stacking these security layers generates immense complexity in computation and hyperparameter tuning [20]. Therefore, before advancing to hybrid architectures, independently measuring the performance trade-offs between the DP pipeline and the SecAgg pipeline serves as a mandatory stepping stone to evaluate the endurance of the network system.

Distributed Differential Privacy

Distributed DP combines the two families: each client adds only a share of the noise, and secure aggregation reveals only the sum, which carries the full calibrated noise [21, 22]. Because secure aggregation operates on integers modulo a large number, discrete noise is preferred; the sum of independent Skellam variables is again Skellam, which permits exact privacy accounting after modular aggregation [22]. The guarantee assumes that enough clients add their share honestly; with fewer honest clients the effective noise is smaller. Distributed DP fits releases that are a single sum, such as count tables, but not multi-step local DP-SGD, where each client's noise passes through non-linear local training before aggregation.

2.1.2 Data Generation

Overview of Data Generation Techniques

In the field of cybersecurity and Intrusion Detection Systems (IDS) for Internet of Things (IoT) environments, Generative AI (GenAI) techniques for data generation are defined as deep generative paradigms capable of learning the underlying multivariate probability distribution of network traffic, thereby automatically synthesizing novel data samples whose structural attributes and statistical properties closely match genuine traffic distributions [23].

Core problems addressed by data generation techniques:

- **Overcoming Extreme Class Imbalance:** Real-world IoT network traffic inherently exhibits extreme class skewness, where over 90–99% of collected traffic consists of benign activity, while malicious behaviors—especially Low-Rate Stealth Attacks (LRSA) or rare DoS/DDoS variants—represent a negligible fraction. This acute shortage of attack samples severely biases downstream IDS classifiers toward majority classes, resulting in unacceptable false-negative rates on critical threats [23].

- **Beyond Interpolation-Based Oversampling:** Traditional statistical oversampling algorithms such as SMOTE [24] operate by performing linear interpolation between existing minority instances within a localized convex hull. As highlighted by Deng et al. [23], deep generative models surpass these rigid boundaries by capturing the global multivariate distribution, enabling the synthesis of samples in unexplored regions of the feature space to cover edge cases and zero-day attack variants, thereby strengthening model robustness. Furthermore, because SMOTE requires simultaneous access to neighboring minority instances in a unified feature space, it is fundamentally ill-suited for federated environments where an individual edge gateway may hold few or no records of a rare attack class [24].

- **Supporting Data Augmentation and Restructuring:** Systematic survey statistics indicate that up to 89% of studies leverage GenAI specifically for data augmentation and class rebalancing, substantially improving the detection performance and generalization of IDS without exposing sensitive raw telemetry [23].

Primary taxonomies of generative architectures in IDS:

- **Generative Adversarial Networks (GANs):** Architectures structured around an adversarial minimax game between a Generator network that creates synthetic samples

from latent noise and a Discriminator network that evaluates sample authenticity. Due to their capacity to model sharp, high-dimensional distributions, GANs account for an overwhelming 58% of research in generative IoT-IDS [23].

- **Autoencoders and Variational Autoencoders (AEs/VAEs):** Frameworks comprising an Encoder that maps high-dimensional traffic into a low-dimensional latent manifold and a Decoder that reconstructs the input. VAEs enforce continuous probability distribution priors over the latent space, permitting stable random sampling to synthesize realistic minority samples [23].

- **Transformer-Based Generative Models:** Architectures leveraging multi-head self-attention mechanisms to capture complex contextual correlations and sequential packet dependencies across temporal network traffic flows [23].

Analysis of Strengths and Weaknesses of Techniques in Existing Studies

Although data generation techniques provide substantial potential for IoT-IDS, critical examination of existing literature reveals distinct computational bottlenecks, training instabilities, and privacy trade-offs when deployed in decentralized Federated Learning (FL) settings [23]:

- **Heavy Generative Denoising Techniques (Tabular Diffusion / Tab-DDPM):** Manzano et al. [14] integrated tabular diffusion models (Tab-DDPM) directly at local FL clients to synthesize minority samples on the BoT-IoT benchmark. While Tab-DDPM achieves exceptional sample fidelity and distribution alignment, the reverse diffusion process requires iterative denoising over hundreds of sequential steps. On edge-tier testbeds (1 CPU core, 2 GB RAM), generating a single synthetic sample requires approximately 32 seconds [14]. Such extreme computational latency is completely unviable for line-rate packet inspection in edge IoT environments.

- **Improved Policy-Guided GANs:** Yang et al. [25] employed an improved GAN guided by reinforcement learning to augment training sets for DDoS detection. While ablation experiments confirm that removing synthetic augmentation causes a noticeable drop in detection performance, the multi-component architecture requires up to 2.726 seconds per batch and consumes 19.6 GB of GPU memory on the KDDcup99 dataset [25].

Furthermore, the study did not isolate standalone generator inference latency, obscuring the true computational efficiency of the generator on edge hardware.

- **Survey Insights on Pure GAN Instability:** In their survey, Deng et al. [23] highlight that while GANs produce high-fidelity samples, the adversarial training dynamics are notoriously brittle and susceptible to mode collapse—wherein the generator repeatedly outputs a narrow set of duplicate patterns, wasting node compute without contributing meaningful diversity.

- **Random Resampling Techniques (Bootstrapping / Oversampling):** Jacob and Sultana [26] utilized bootstrapping (random resampling with replacement) at local nodes to balance the BoT-IoT and CSE-CIC-IDS2018 datasets. While computationally trivial and easily executable on microcontrollers, bootstrapping merely duplicates existing data points without expanding feature diversity. This encourages classifier overfitting and fails to generalize against evolving attack variants or concept drift [23, 26].

- **Cloud-Based Centralized Data Generation Techniques (Cloud-Based VAE-GAN):** Anandhi and Srikanth [27] proposed the SSAIM framework utilizing a hybrid VAE-GAN at the cloud coordinator to generate 51,200 Low-Rate Stealth Attack (LRSA) samples for the IoTID20 dataset, raising attack recall from 62% to 98%. However, placing data synthesis in the cloud violates the decentralized privacy principle of FL. Additionally, compressing the heavy VAE-GAN model via fixed-point quantization for transmission back to edge devices demonstrated that quantization below 6-bit (e.g., 4-bit) caused a severe degradation in detection sensitivity [27].

- **Federated Lightweight Edge GANs (TinyGAN in FL):** Durga et al. [28] deployed TinyGAN within a federated framework for IoMT malware detection, reporting 99.30% accuracy after 20 rounds. However, to stabilize GAN training, all malware classes representing less than 0.5% of the dataset were discarded, thereby sidestepping the core challenge of extreme class imbalance. Moreover, the study omitted standalone sample generation throughput measurements, leaving its actual computational advantage on constrained edge hardware unproven [28].

2.1.3 Privacy-Preserving Generative Defenses and Research Gap

Privacy-Utility Dilemma in Federated Generative IDS

Federated intrusion detection systems (IDS) on edge networks must protect client telemetry while coping with severe minority-attack-class imbalance and non-IID traffic distributions. Differential privacy (DP) and cryptographic protocols bound gradient inversion and membership inference risks [29] but do not correct class skew. Augmenting under-represented attack classes with federated generative models mitigates this imbalance, yet generative architectures may memorize training records and expose sensitive network traces through parameter exchanges or synthesized flows [29]. Edge defenses must therefore limit the utility loss of synthetic data while providing formal client-side privacy guarantees.

Existing privacy-preserving generative IDS frameworks use cryptographic primitives, DP, or both. Cryptographic mechanisms, such as secure multi-party computation (MPC) and threshold secret sharing, preserve numerical precision without perturbation error [29]. However, secure aggregation of high-dimensional neural parameters requires iterative multi-round communication, which burdens resource-constrained IoT edge devices [29].

Empirical Generative PPFL-IDS and Cryptographic Limitations

A literature search of IEEE Xplore and SpringerLink (2022–2026) for studies combining federated learning, intrusion detection, and generative models (GAN or VAE) identified five empirical implementations, all from IEEE [30–34]. None applies cryptographic secure aggregation to uplink generator updates. Salim et al. [32] (global VAE aggregation), Jiao et al. [33] (local CGAN gradients), and Shettar et al. [34] (central aggregator with moments-accountant clipping at $\varepsilon = 4.8$ and $\delta = 10^{-5}$) rely on DP alone. He et al. [31] combine local gradient noise with blockchain Proof-of-Authority consensus across edge servers, which ensures aggregation integrity and auditability rather than parameter confidentiality.

Yao et al. [30] use a hybrid design: proactive dynamic (t,n)-threshold secret sharing with $t = 4$ and $n = 5$ protects only downlink synthetic broadcasts, whereas uplink client gradients rely on Gaussian DP. They report execution measurements for key distribution and sample reconstruction (Table VI), but secret sharing covers downlink tabular samples, not full model-parameter aggregation. Outside the primary search, Hamouda et al. [35] likewise omit secure aggregation and identify cryptographic model protection as an open challenge. Because

none of these works benchmarks cryptographic aggregation against DP on model weights, their use of DP reflects a design choice rather than a measured trade-off.

The effect of DP perturbation on downstream detection depends on the privacy budget ε, gradient clipping threshold C, and minority-class sample density, yet existing studies do not isolate these factors. Utility can remain high at a moderate budget or with large synthetic volumes: Shettar et al. [34] report a 96.8% detection F1-score under central DP ($\varepsilon = 4.8$), and Hamouda et al. [35] observe a 0.25-percentage-point accuracy reduction at $\varepsilon = 1.0$ when applying DP-SGD to the downstream global classifier with 50,000 synthetic samples per class. Generative augmentation can also raise rare-class detection from sparse baselines: He et al. [31] report that CGAN augmentation improved Infiltration F1-score from 33.33% to as high as 98.01%, starting from 29 baseline training samples. However, Hamouda et al. [35] applied DP to the classifier rather than the generator, and neither He et al. [31] nor Jiao et al. [33] report formal numerical (ε, δ) budgets. Because generator-side DP must spread privacy noise across iterative training steps, how strict bounds ($\varepsilon \leq 1.0$) interact with extreme class scarcity in federated generative architectures remains an open empirical question.

Marginal-Based Tabular Synthesis vs. Deep Generative Models

Marginal-based tabular synthesis offers an alternative to deep generative models. Instead of optimizing non-convex neural network weights by backpropagation, marginal synthesizers select and measure low-order contingency tables and reconstruct the data distribution with probabilistic graphical models. Beyond network IDS research, a cross-domain search of the ACM Digital Library, PMLR, and arXiv identified distributed extensions: Maddock et al. [36] introduce FLAIM, which adapts the Adaptive and Iterative Mechanism (AIM) to distributed clients under zero-concentrated differential privacy (ρ-zCDP) and compares DP with secure multi-party computation (DistAIM); Pentyala et al. [37] present CaPS, a "DP-in-MPC" framework in which secret sharing protects marginal selection and measurement; and Li et al. [38] propose HeteroFedSyn, which compresses two-way marginal dependencies by random projection under distributed perturbation.

Marginal synthesizers avoid the adversarial minimax training instability of GANs but carry their own privacy-utility trade-offs. Iterative selection-and-measurement algorithms (e.g., AIM and Private-PGM) divide the total privacy budget ε across rounds, and distributing measurements across federated clients compounds perturbation noise [36, 37]. In sparse discrete categories with small contingency cell counts, additive Laplace or Gaussian noise

yields a low signal-to-noise ratio. Restricting measurements to low-order marginals (typically two-way dependencies [38]) truncates higher-order multivariate correlations, and continuous flow variables require discretization, which introduces binning artifacts.

Federated marginal generators have so far been evaluated mainly on general-purpose tabular benchmarks from demographic, financial, and biomedical domains (e.g., Adult, Census, Credit Card Fraud, COMPAS, Diabetes) [36–38]. FLAIM [36] also used the legacy DARPA Intrusion Detection benchmark but evaluated only workload error and negative log-likelihood, not downstream minority-class intrusion classification. Decentralized tabular GANs with cryptographic primitives also exist: Duan et al. [39] combine Paillier additively homomorphic encryption for global multimodal distribution encoding with DP during GAN training, but evaluate statistical utility and membership inference resistance on legacy tabular data (KDD Cup 99), not modern protocol telemetry.

IoT-MQTT Protocol Telemetry and Working Hypotheses

IoT-MQTT telemetry combines structured, low-cardinality discrete protocol fields (e.g., message types, QoS levels, retain flags) with continuous flow metrics (e.g., payload lengths, burst durations). Sparse discrete feature combinations form low-count contingency cells, where both generative families risk degradation under DP noise, making this a suitable setting for examining these trade-offs. We hypothesize that, under generator-side DP at strict budgets ($\varepsilon \leq 1.0$), a federated marginal synthesizer achieves higher minority-class detection F1-score than a federated deep generative model for attack classes defined by low-order (two-way) discrete header correlations (e.g., MQTT Unauthorized Connect, Malformed Header). We expect the reverse for volumetric or timing-based attacks (e.g., MQTT Publish Floods) governed by multi-attribute continuous flow dynamics. We test this across five random seeds, equalizing privacy budgets by converting all mechanisms to (ε, δ)-DP under Rényi differential privacy accounting, and compare class-specific F1-scores with a Wilcoxon signed-rank test (treating a margin $|\Delta\text{F1}| \geq 2.0$ percentage points at $p < 0.05$ as statistically significant).

Identified Research Gaps

To the best of our knowledge, the IEEE, Springer, ACM, PMLR, and arXiv searches described above indicate three distinct research gaps:

- **Marginal Synthesis for Protocol Telemetry:** Deep generative models have been evaluated on MQTT-bearing datasets (e.g., Hamouda et al. [35], federated conditional GAN on Edge-IIoTset), but no prior study evaluates a federated marginal-based generator on modern IoT-MQTT telemetry or examines how mixed discrete-continuous features affect marginal synthesis relative to deep models.
- **Controlled Architectural Comparison:** No empirical study compares a federated marginal-based synthesizer with a federated deep generative model (e.g., CGAN, VAE) under identical non-IID client partitions, minority-attack imbalance ratios, and calibrated generator-side privacy budgets (ε, δ).
- **Downstream Detection Trade-offs:** Prior marginal studies compared DP with cryptographic aggregation on workload error, negative log-likelihood, runtime, and communication bandwidth [36, 37], and decentralized GANs were evaluated for statistical utility and membership inference resistance [39]. No study quantifies how choosing between DP and cryptographic secure aggregation affects downstream minority-class intrusion detection under matched federated edge conditions.

This study addresses these gaps by evaluating communication overhead, synthetic sample fidelity, and downstream minority-class intrusion detection for marginal and deep generative models under both DP and cryptographic secure aggregation on edge IoT networks.

2.2 Summary of the Literature Review

2.3 Contribution of Research

Chapter 3

Methodology

Describes the research design, data collection methods, and analysis techniques.

3.1 Research Design

3.2 Data Collection Methods

3.3 Sampling Data Analysis Techniques

3.4 Limitations of the Methodology

Chapter 4

Experimental and Results

Provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn. You should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses.

4.1 Introduction

4.2 Presentation of Data

4.3 Analysis of Results

4.4 Interpretation of Results

4.5 Comparison with Literature

4.6 Implications of the Results

Chapter 5

Discussion

The findings and their implications should be discussed in the broadest context possible and limitations of the work highlighted.

5.1 Restate the Research Problem or Objectives

5.2 Summarize Key Findings

Chapter 6

Conclusion and Future Work

The 'conclusions' are a key component of the paper. It should complement the 'abstract' and is normally used by experts to value the paper's engineering content. A conclusion is not merely a summary of the main topics covered or a re-statement of your research problem, but a synthesis of key points and, if applicable, where you recommend new areas for future research.

6.1 Conclusion

6.2 Future Work

Appendix A

If Any

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